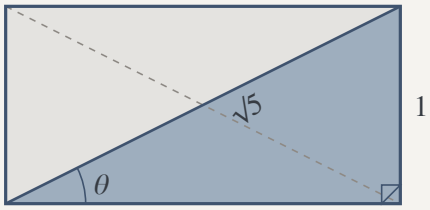
THE PINWHEEL TILING

a pattern that never repeats, made from ordinary 2 : 1 subway tile

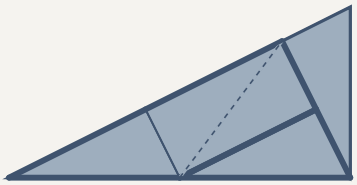
I. ONE TILE, ONE CUT



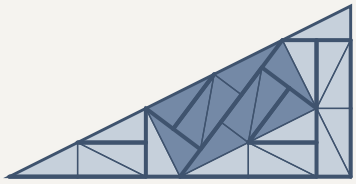
cut along either diagonal

Most aperiodic tilings are impractical to build. Penrose tiles need golden-ratio proportions and 36° and 72° angles, so they must be custom-made. The pinwheel tiling needs only a standard 3″ × 6″ subway tile, cut once along a diagonal; the triangle's smallest angle is $\theta = \arctan \frac{1}{2} \approx 26.565°$. There are two diagonals, and cutting along each yields different triangles: left-handed and right-handed. Two same-handed triangles meeting along their hypotenuse pair back up into full tiles; differently-handed triangles meeting along their hypotenuse become kites.[†]

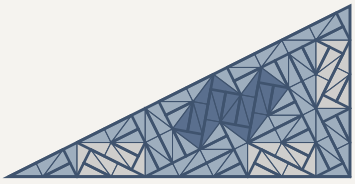
II. FIVE FROM ONE



one generation: 5 tiles, all tilted ■ θ



two: 25, tilted ■ 0 or ■ 2θ



three: 125, tilted ■ −θ, ■ θ or ■ 3θ

The 2:1 triangle divides exactly into five copies of itself, each scaled by $1/\sqrt{5}$: the altitude from the right angle splits off one, and the rest divides into four, two of which form an uncut subway tile.[‡] Enlarge by $\sqrt{5}$ and repeat: 1 tile becomes 5, then 25, then 125, covering ever larger regions of the plane. This tiling was discovered by John Conway. Charles Radin (1994) showed that this tiling can be enforced with decorated copies of the tile and local matching rules.

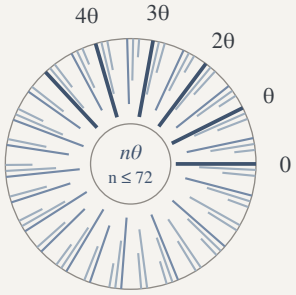
III. THE IRRATIONAL TWIST

1. Each child's long leg (marked in bold above) is tilted by exactly θ from its parent's; perpendicular directions count as the same. Three children are mirror images of the parent; two are not.

2. A mirrored tile's children turn the opposite way from an unmirrored tile's, so each generation adds $\pm\theta$. After n generations tiles thus point in directions $-(n-2)\theta$, $-(n-4)\theta$, …, $(n-2)\theta$, $n\theta$.

3. A periodic tiling is one finite patch repeated forever, so it can only have finitely many directions. As long as the directions $k\theta$ are all distinct, the tiling must be aperiodic. (If no child were mirrored, every tile would point in direction $n\theta$, and the directions would be finite!)

4. $\arctan \frac{1}{2}$ is not a rational multiple of π,[§] so no two directions $k\theta$ coincide, even up to a multiple of 90°. A pinwheel tiling contains supertiles of every generation, so it has infinitely many directions (uniformly distributed, as Radin showed). **Therefore no pinwheel tiling is periodic.**



directions nθ never repeat

It never repeats because $\arctan \frac{1}{2}$ is not a rational multiple of π.

[†] *For the attentive viewer:* the tiles are exactly 2 : 1, but the grout joints are not uniform. For tile plus grout to stay on a 2 : 1 grid, $6'' + g_{end} = 2(3'' + g_{side})$, so $g_{end} = 2g_{side}$: ⅛″ joints along the long sides, but ¼″ at the short ends. Can you see the difference?

[‡] The same works for any N : 1 tile. The altitude cuts off one copy; the rest is the triangle enlarged N times, which divides into N^2 copies (N along the hypotenuse and $N(N-1)/2$ whole tiles), so the triangle divides into $N^2 + 1$ copies scaled by $1/\sqrt{(N^2+1)}$. For each, $\theta = \arctan(1/N)$: a 4″ × 12″ tile gives a 10-piece pinwheel, a 2″ × 8″ tile a 17-piece one. $N = 1$ is degenerate and periodic, since $\arctan 1 = \pi/4$ *is* a rational multiple of π.

[§] Since $\tan\theta = \frac{1}{2}$, $2 + i = \sqrt{5}\, e^{i\theta}$. If $\theta = (p/q)\pi$, then $(2+i)^{2q} = 5^q = (2+i)^q(2-i)^q$, so $(2+i)^q = (2-i)^q$, impossible since $2 \pm i$ are distinct Gaussian primes. More generally, the only rational values of tan at rational multiples of π are 0 and ±1 (I. Niven, *Irrational Numbers*, 1956). The irrationality of π itself is beside the point: $\arctan 1 = \pi/4$ is a rational multiple of π, and right isosceles triangles tile periodically.

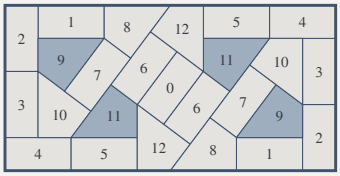
C. Radin, "The pinwheel tilings of the plane," Ann. of Math. 139 (1994), 661–702. Border: this wall's tiling, extended, 1 : 24.

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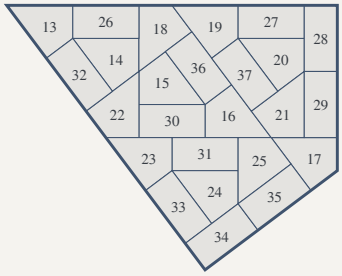
COLORING THE PINWHEEL

dominoes, kites, and a 38-bit mask

IV. DOMINOES AND KITES



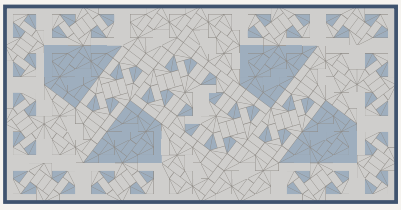
domino → 15 dominoes + 10 kites



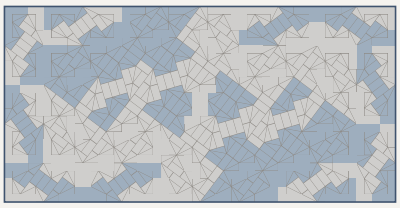
kite → 12 dominoes + 13 kites

On the wall, triangles are paired across their hypotenuses: a *domino* is a whole 3″ × 6″ tile, a *kite* is two mirror-image half-tiles. Taking two pinwheel generations at once gives a substitution that inflates by 5, shown at left. Children of a domino are numbered 0–12 (pieces swapped by its half-turn symmetry share a number); children of a kite, 13–37.

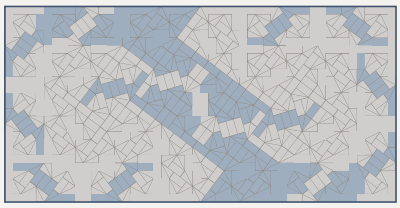
A coloring is a set S of these numbers. Each piece is one color step from its parent if its number is in S, and the same color otherwise, counting modulo the number of colors. With the two tiles on this wall, $S = \{9, 11\}$, the two shaded kites in every domino, gives the border around this page. Below are other sets explored while designing the wall, each on a level-2 domino of 625 pieces, with their original working names.



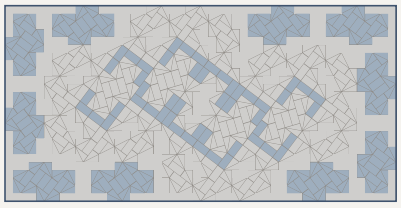
kites (this wall) {9, 11}



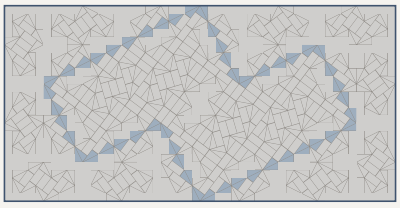
s hook {0, 1, 2, 6, 7, 8, 9}



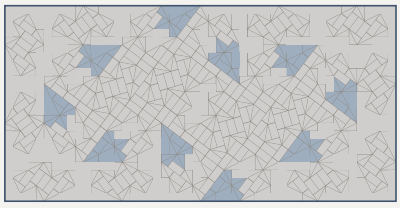
whip {0, 1, 6, 8}



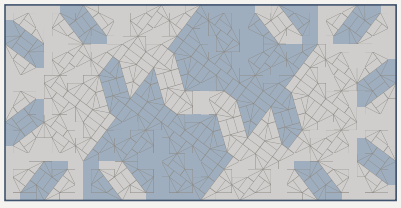
triple block outline {1, 2, 3, 4, 5}



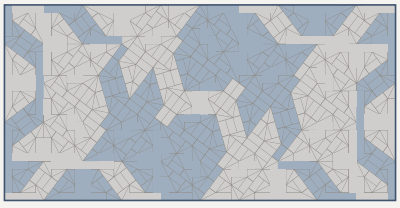
tetris Z piece {13, 14, 15, 16, 17}



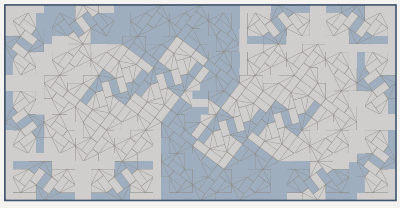
rocket ships {13, 14, 15, 18, 22, 26, 32}



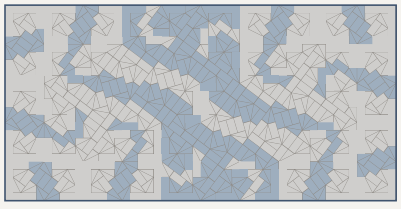
simple worms (best) {0, 5, 6, 11, 12}



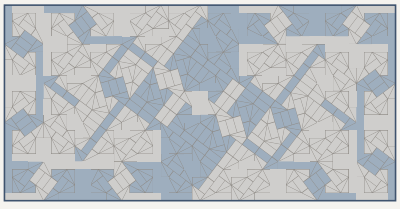
bracket worms {0, 4, 5, 6, 11, 12}



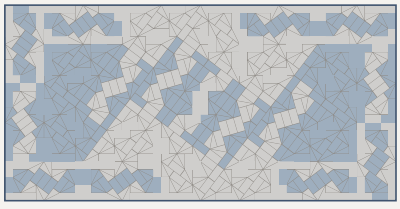
z snake {0, 1, 8, 12}



worms 2e {0, 6, 8, 12, 16, 19, 23, 31, 37}



worms 3 {0, 4, 5, 6, 12, 17, 34, 35}



samson's columns {0, 3, 6, 7, 9, 10}

Colorings from github.com/cscott/tile-hack, in this wall's tiles: weather gray and skylight blue.

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